

Measured and source-confirmed facts

96 of 96
workers active

Input lock
occupied 19.72
s
49.52% of
39.82 s
mapping interval

Aggregate
mapping
concurrency
71.28
equivalent CPUs

Active-worker
finish spread
12.93 s

Parsing occurs
inside the
shared
critical section
in release
source

A02 bounded implementation gate

Corrected full-
corpus timing
diagnostic
complete

Which
candidate
reduces
corrected wall
time?

Faster record-
safe input
preparation

Finer
scheduling units
SELECTED
FOR A02

Bounded
producer /
consumer queue

No qualifying
gain
rollback and
archive

Measured
opportunity, not
achieved speed:
stages overlap
and bottlenecks
can move

One logical
FASTQ stream
per mate

Shared
mutexInRead

One worker
parses and
copies
a complete
input chunk
while locked

Release input
lock

That worker
maps its private
chunk

Another worker
may acquire
input lock

